

SKILLFORGE

Project Synopsis

1. Introduction

Choosing a suitable career path can be difficult for students entering fast-changing technology fields. Many students have basic technical knowledge but lack clear guidance on the exact skills, tools, certifications, and practical experience required for a particular job role. SkillForge is designed to provide continuous, skill-based career guidance through one integrated web application.

2. Problem Statement

Traditional career guidance is often generic, one-time, or dependent on manual counselling. Static career listings and standalone learning resources do not clearly connect a student's existing skills with career requirements, skill gaps, learning roadmaps, and measurable progress. A system is therefore needed to provide personalized and structured career preparation based on a user's current profile and skills.

3. Objectives

- Provide personalized career recommendations based on user skills and profile.
- Compare current skills with career requirements and identify skill gaps.
- Generate a structured, career-specific learning roadmap.
- Recommend relevant projects and certifications for career preparation.
- Track roadmap completion and present progress in an understandable form.
- Provide administrative facilities for managing system data and monitoring progress.

4. Proposed System

SkillForge connects profile management, skills, careers, skill-gap analysis, roadmaps, projects, certifications, recommendations, and progress tracking into one data-driven workflow. The system calculates a career match percentage by comparing the user's saved skills with the required skills defined for each career. The result can then be used to support career selection, skill-gap identification, roadmap planning, and progress monitoring.

5. System Flow

Authentication → Profile → My Skills → Career Recommendation → Skill Gap Analysis → Project & Certification Recommendation → Personalized Career Roadmap → Progress Tracking → Dashboard & Reports

6. Modules

1. User Authentication & Profile Management

Handles user registration, login, profile information, educational details, profile updates, and account status.

2. Skills Management

Allows users to add, view, manage, and remove skills. These skills form the basis for career matching and skill-gap analysis.

3. Career Recommendation

Compares a user's saved skills with career requirements and generates career match percentages. Preferred careers can be selected and saved.

4. Skill Gap Analysis

Identifies missing or insufficient skills by comparing the user's current skill set with the requirements of a selected career.

5. Personalized Career Roadmap

Provides a structured learning path with sequenced roadmap steps, learning activities, and resources for the selected career.

6. Project & Certification Recommendation

Suggests relevant projects and certifications that help users strengthen practical skills and career readiness.

7. Progress Tracking

Tracks roadmap completion, status, and progress percentage as users complete learning activities.

8. Dashboard & Reports

Provides a centralized view of profile, skills, recommendations, skill gaps, roadmap summaries, and progress using cards, indicators, and charts.

7. Database Design

The planned normalized database contains 10 main tables. The core relationship model links users with profiles and skills, careers with required skills, recommendations with users and careers, roadmaps with careers, and progress with roadmap activities.

Table	Purpose
auth_user	Stores user registration, login, password, and account-status information.
core_profile	Stores user profile and educational details.
core_skill	Stores the skills available in the system.
core_userskill	Links users with their selected skills.
core_career	Stores career information.
core_careerskill	Maps required skills to careers and records skill importance.
core_roadmap	Stores step-by-step learning roadmap entries for careers.

Table	Purpose
core_recommendation	Stores career recommendations and match scores for users.
core_progress	Stores roadmap progress, status, and completion percentage.
core_project_certification	Stores career-related project and certification recommendations.

8. Technology Stack

Backend: Python and Django

Frontend: HTML, CSS, Bootstrap and JavaScript where required

Database: MySQL/MariaDB through XAMPP and phpMyAdmin in the current development setup

Visualization: Chart.js planned for dashboard and report visualizations

Development Environment: Python 3.12.10, Django 4.2.27, Visual Studio Code

9. Student-Side Functional Flow

The student-side system begins with authentication. After login, the student manages their profile and skills, views career recommendations, analyzes skill gaps, explores career-related projects and certifications, follows a personalized roadmap, and tracks progress. The dashboard summarizes information generated by these functions.

10. Administrative Functions

Administrative functionality is provided for platform management rather than as separate student modules. Main administrative functions include User Management, Skill Management, Career Management, Roadmap Management, Project & Certification Management, and Progress Monitoring.

11. Expected Outcome

The expected outcome is a single platform that helps students move from their current skill profile to a clearer career-development path. The system provides career matching, skill-gap identification, learning-roadmap support, project and certification guidance, and measurable progress tracking in an integrated workflow.

12. Conclusion

SkillForge is proposed as a personalized career-planning system that combines skill-based career recommendation with structured learning and progress tracking. By connecting career requirements to user skills, learning roadmaps, projects, certifications, and progress, the system aims to support students in making informed career-development plans.

Note on current development state

The current implementation has completed the core authentication, profile, skills, and initial career-recommendation development stages. Career and CareerSkill data population and subsequent modules are being developed incrementally.